Lab Report: Image classification experiments

1 Introduction

In this report we compare several models on standard benchmarks. The dataset was split into 70 percent training data and the rest for testing. All experiments were run on a single GPU with 24 GB of memory.

2 Results

Figure 1 shows the f1 score during training.

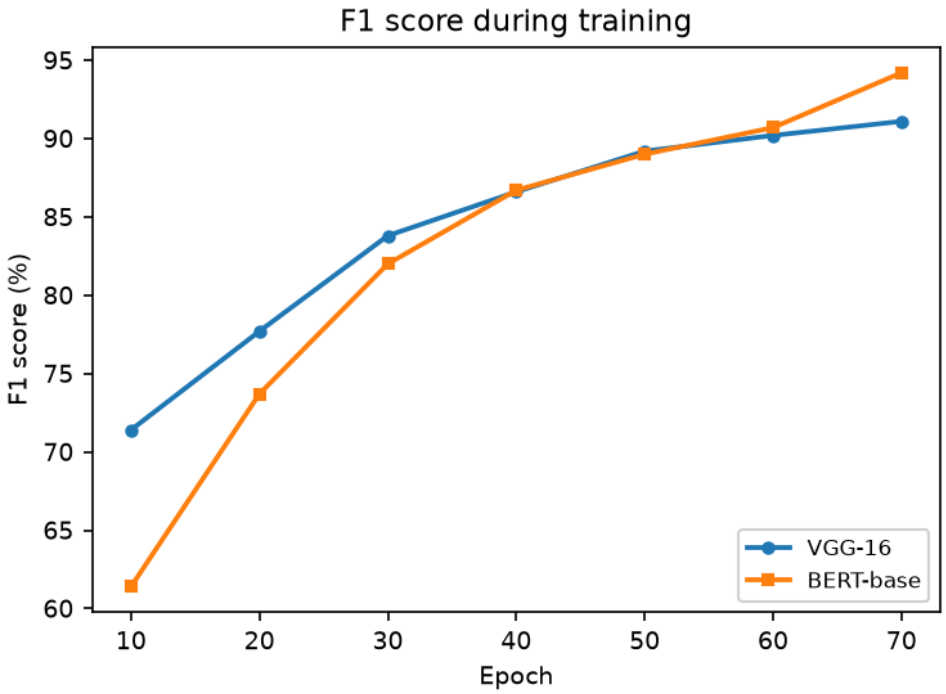


Figure 1: F1 score during training for the models in this experiment.

BERT-base attains a F1 score of about 90% at epoch 70. VGG-16 reaches 104.6% F1 score at epoch 40. At epoch 50, VGG-16 achieves 89.2% F1 score.

Figure 2 shows the accuracy of MobileNet by dataset.

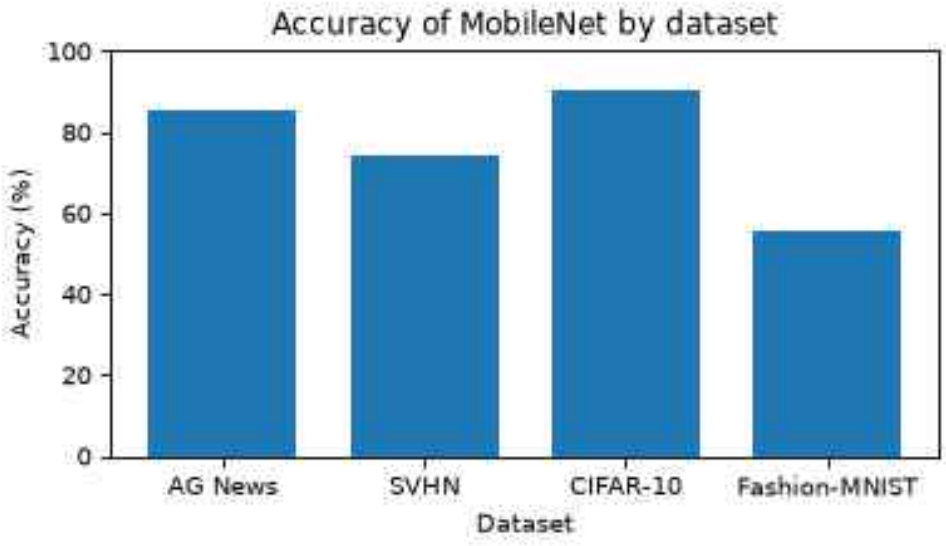


Figure 2: Accuracy of MobileNet by dataset for the models in this experiment.

MobileNet reaches 53.9% accuracy on Fashion-MNIST. As shown in Figure 2, MobileNet achieves 85.5% accuracy on AG News. MobileNet scores 4.6 points higher on CIFAR-10 than on AG News.

Figure 3 shows the validation accuracy during training.

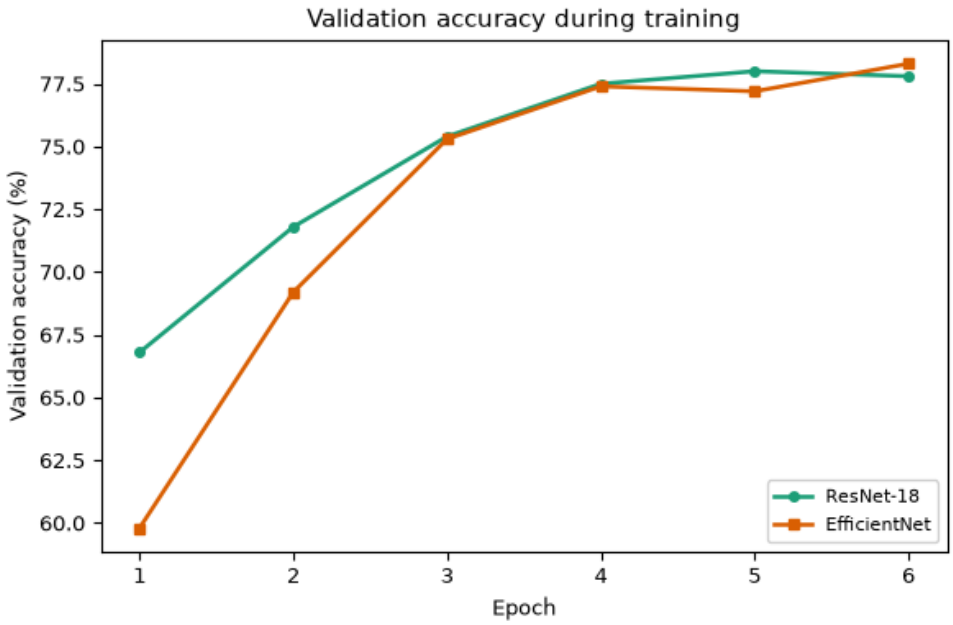


Figure 3: Validation accuracy during training for the models in this experiment.

At epoch 3, ResNet-18 achieves about 80% validation accuracy. As shown in Figure 3, ResNet-18 reaches 79.0% validation accuracy at epoch 5. EfficientNet reaches 78.3% validation accuracy at epoch 6 (Figure 3). ResNet-18 attains a validation accuracy of 77.5% at epoch 4.

Figure 4 shows the test accuracy during training.

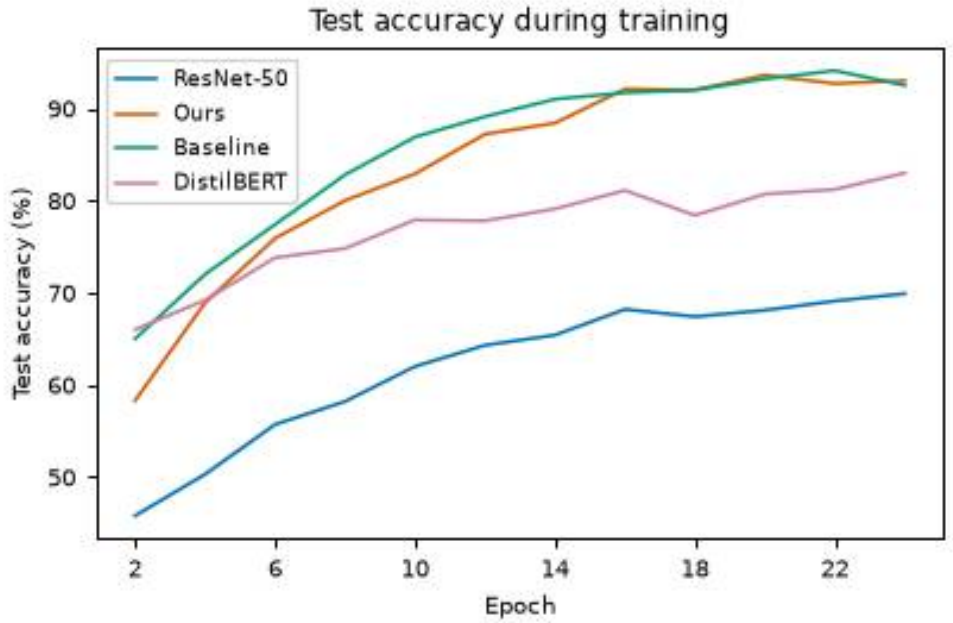


Figure 4: Test accuracy during training for the models in this experiment.

From epoch 2 to epoch 24, the test accuracy of Baseline improves by a relative 50.5%. As shown in Figure 4, from epoch 22 to epoch 24, the test accuracy of DistilBERT improves by a relative 8.8%. As shown in Figure 4, DistilBERT attains a test accuracy of 81% at epoch 22. From epoch 14 to epoch 16, the test accuracy of DistilBERT improves by a relative 9.2%.

3 Discussion

On Fashion-MNIST, Naive Bayes obtains an accuracy of 56.2%. We note that the training was stable for all models.